Stable Sampling for Riesz–Feller Bernstein Spaces

Explicit RKHS kernels, spacetime reconstruction, and endpoint obstructions

Candidate substantial partial result for the outlook of arXiv:2405.04989

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Status: candidate substantial partial result, likely valid. We give the complete $p = 2$ RKHS and a stable Whittaker–Shannon reconstruction of the entire Riesz–Feller spacetime solution from one lattice of boundary samples. Two exact bandlimited examples prove that the same-space Hardy splitting fails at both $p = 1$ and $p = \infty$; a bandpass condition repairs the obstruction. The general Hardy/BMO endpoint theory remains open.

1 Source outlook and result

Bernstein and Faustino identify their generalized Bernstein space with Clifford-valued functions whose ordinary Fourier transform is supported in a ball, then leave the following two directions on source PDF p. 29 [1].

6 Some Outlook

The foundations of a Paley-Wiener theory have been developed, and this theory accurately describes Clifford-valued functions of exponential type R^α, $\boldsymbol{u}$, from the knowledge of the support of the Fourier transform of $\boldsymbol{f} = \boldsymbol{u}(\cdot, 0)$, and vice versa. In particular, the spectral formula for the maximum radius R, provided by the Theorem 7, allows for an explicit calculation of the so-called bandwidth of the generalized Bernstein space $\mathcal{B}^p_R(\boldsymbol{D}^\alpha_\theta)$.

The predominant comprehension of $\mathcal{B}^p_R(\boldsymbol{D}^\alpha_\theta)$ indicates the possibility for a hypercomplex extension of the Whittaker-Shannon sampling theorem, analogous to the one outlined in [29]. However, further investigation is necessary to develop numerical implementations of sampling reconstruction schemes. This investigation necessitates a comprehensive understanding of *R*eproducing Kernel Hilbert Spaces, which is beyond the scope of the present paper. The authors intend to explore these topics in a future publication.

In a different direction, it would be worthwhile to consider the potential applications of our results for values of $0 < p \leq 1$ and $p = \infty$. While many steps in our proofs appear to necessitate only minor alterations, as evidenced by the results obtained in [14], this approach is contingent on intricate properties involving duality theorems between Hardy-type spaces H^1 and BMO-type spaces, as well as, more generally, between Hardy-type spaces H^p and Morrey-Campanato type spaces (see, for example, [16, 57] for a comprehensive overview).

Exact transcription of the live passages. "The predominant comprehension of $\mathcal{B}^p_R(D^\alpha_\theta)$ indicates the possibility for a hypercomplex extension of the Whittaker-Shannon sampling theorem, analogous to the one outlined in [29]. However, further investigation is necessary to develop numerical implementations of sampling reconstruction schemes. This investigation necessitates a comprehensive understanding of Reproducing Kernel Hilbert Spaces, which is beyond the scope of the present paper." They then write: "In a different direction, it would be worthwhile to consider the potential applications of our results for values of $0 < p \leq 1$ and $p = \infty$."

We completely answer the Hilbert-space sampling/RKHS part, including stable finite reconstruction, and prove that a naive endpoint extension is false. Clifford-valued ball sampling itself is classical [2]; the new content relative to the outlook is the explicit stable propagation through the Riesz–Feller multiplier and the endpoint diagnosis.

Proof Intuition

At $p = 2$, the difficult analytic work has already been encoded by the source's Fourier-support theorem. A function supported in a frequency ball can be extended by zero to a containing cube. Its Fourier-series coefficients on that cube are exactly its samples on the reciprocal lattice. Applying the Riesz–Feller solution multiplier term by term produces a spacetime sampling kernel. Parseval supplies exact sample stability, while compact frequency support turns L^2 convergence into uniform convergence.

The endpoint issue is concentrated at frequency zero. The Riesz–Hilbert multiplier jumps there. The transforms of $(\sin x/x)^2$ and of the bounded sine-integral function expose, respectively, a $1/x$ tail and logarithmic growth. Removing a neighborhood of zero makes the multiplier smooth and restores both endpoint bounds by Young's inequality.

2 The source multiplier and the RKHS

Let $V = \mathrm{Cl}_{0,n}^{\mathbb{C}}$ with its finite-dimensional coefficient Hilbert norm. We use the source convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-ix\cdot\xi}\,dx, \qquad f(x) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \widehat{f}(\xi)e^{ix\cdot\xi}\,d\xi.$$

For $\xi \neq 0$, put

$$\chi_{\pm}(\xi) = \frac{1}{2}\left(1 \pm \frac{i\xi}{|\xi|}\right).$$

The source solution with boundary datum f is $u_f(x,t) = \mathcal{F}^{-1}(M_t\widehat{f})(x)$, where

$$M_t(\xi) = \begin{cases} e^{-t|\xi|^\alpha}\chi_-(\xi), & t > 0,\\ 1, & t = 0,\\ e^{-te^{-i\pi\theta}|\xi|^\alpha}\chi_+(\xi), & t < 0. \end{cases} \tag{1}$$

Here $0 < \alpha \le 1$ and $|1-\theta| < \alpha/2$. Thus $\cos(\pi\theta) < 0$, both scalar factors in (1) have modulus at most one in their stated half-spaces, and

$$\sup_{t\in\mathbb{R},\,\xi\neq 0} \|L_{M_t(\xi)}\|_{\mathrm{op}} \le C_\chi, \qquad C_\chi := \max\{1, \sup_\xi \|L_{\chi_\pm(\xi)}\|_{\mathrm{op}}\} < \infty. \tag{2}$$

The letter L denotes left Clifford multiplication.

Let

$$\mathrm{PW}_R^2(V) = \{f \in L^2(\mathbb{R}^n; V) : \operatorname{supp}\widehat{f} \subseteq \overline{B(0,R)}\}.$$

The source theorem identifies these data with its $p = 2$ generalized Bernstein solutions.

Theorem 1 (Boundary and spacetime RKHS)**.** *The boundary space $\mathrm{PW}_R^2(V)$ is a V-valued RKHS with kernel*

$$\begin{aligned} k_R(x,y) &= \frac{1}{(2\pi)^n}\int_{B(0,R)} e^{i(x-y)\cdot\xi}\,d\xi\, I_V \\ &= \frac{R^{n/2}}{(2\pi)^{n/2}|x-y|^{n/2}} J_{n/2}(R|x-y|)I_V, \end{aligned} \tag{3}$$

with diagonal value $|B(0,R)|/(2\pi)^n$.

Norm the solution space $\mathcal{H}_R = \{u_f : f \in \mathrm{PW}_R^2(V)\}$ by $\|u_f\|_{\mathcal{H}_R} = \|f\|_2$. It is an operator-valued RKHS on $\mathbb{R}^n \times \mathbb{R}$ with kernel

$$\mathbb{K}((x,t),(y,s)) = \frac{1}{(2\pi)^n}\int_{B(0,R)} e^{i(x-y)\cdot\xi} L_{M_t(\xi)}L^*_{M_s(\xi)}\,d\xi. \tag{4}$$

At $t = s = 0$, (4) reduces to (3).

Proof. Cauchy–Schwarz on the finite-measure ball makes every boundary evaluation bounded. Fourier inversion of the ball indicator gives (3); the displayed Bessel formula follows by polar coordinates and has the stated continuous value at zero.

For $z = (x,t)$, define

$$E_z\widehat{f} = \frac{1}{(2\pi)^n}\int_{B(0,R)} e^{ix\cdot\xi}L_{M_t(\xi)}\widehat{f}(\xi)\,d\xi.$$

Equation (2) makes E_z bounded. Direct computation gives $E_zE_w^*$ equal to (4); hence that kernel is positive and reproducing. Since $u_f(\cdot,0) = f$, the boundary norm defines a genuine Hilbert norm on the solution space. □

3 Exact stable sampling of the full solution

Write $\operatorname{sinc} z = \sin z/z$, with $\operatorname{sinc} 0 = 1$.

Theorem 2 (Spacetime Whittaker–Shannon formula)**.** *Let $\Omega \geq R$, $a = \pi/\Omega$, and $Q_\Omega = [-\Omega, \Omega]^n$. For $f \in \mathrm{PW}^2_R(V)$ and its source solution u_f, define*

$$\Psi_{k,t}(x) = \frac{a^n}{(2\pi)^n} \int_{Q_\Omega} M_t(\xi) e^{i(x-ak)\cdot\xi}\, d\xi, \qquad k \in \mathbb{Z}^n. \tag{5}$$

Then

$$u_f(x,t) = \sum_{k\in\mathbb{Z}^n} \Psi_{k,t}(x) f(ak). \tag{6}$$

The series converges in slice L^2, uniformly in (x,t), and at the boundary

$$\Psi_{k,0}(x) = \prod_{j=1}^{n} \operatorname{sinc}(\Omega x_j - \pi k_j). \tag{7}$$

Moreover the sample map is an exact scaled isometry:

$$\|f\|_2^2 = a^n \sum_{k\in\mathbb{Z}^n} |f(ak)|_0^2. \tag{8}$$

For a finite $\Lambda \subset \mathbb{Z}^n$, let u_Λ be the corresponding partial sum and $\tau_\Lambda = (\sum_{k\notin\Lambda} |f(ak)|_0^2)^{1/2}$. Then

$$\sup_t \|u_f(\cdot,t) - u_\Lambda(\cdot,t)\|_2 \leq C_\chi a^{n/2} \tau_\Lambda, \qquad \sup_{x,t} |u_f(x,t) - u_\Lambda(x,t)|_0 \leq C_\chi \tau_\Lambda. \tag{9}$$

The same bounds hold with τ_Λ replaced by the ℓ^2 norm of additive sample noise.

Proof. Extend $\widehat{f}$ by zero from $B(0,R)$ to Q_Ω. Its V-valued Fourier series is

$$\widehat{f}(\xi) = a^n \sum_{k\in\mathbb{Z}^n} f(ak) e^{-iak\cdot\xi} \quad \text{in } L^2(Q_\Omega; V). \tag{10}$$

Indeed the Fourier coefficient is $a^n f(ak)$ by Fourier inversion. Multiplying (10) on the left by M_t and inverting gives (6); this also explains why the Clifford sample stays on the right of (5). Setting $t = 0$ and integrating each coordinate gives (7). Fourier-series Parseval followed by Euclidean Plancherel gives (8) with no hidden constant.

If f_Λ is the boundary partial sum, Parseval gives $\|f - f_\Lambda\|_2 = a^{n/2}\tau_\Lambda$. Equation (2) proves the first estimate in (9). Every propagated error is Q_Ω-bandlimited, so Cauchy–Schwarz in Fourier space gives

$$\|g\|_\infty \leq \left(\frac{|Q_\Omega|}{(2\pi)^n}\right)^{1/2} \|g\|_2 = a^{-n/2}\|g\|_2.$$

This cancels the preceding $a^{n/2}$ and proves the uniform estimate. The same linear argument applies to any ℓ^2 noise sequence and also proves the asserted convergence. □

Thus finite boxes Λ provide a directly implementable stable scheme: precompute the kernels (5), acquire boundary samples, and certify the error from the omitted or noisy sample ℓ^2 norm.

4 Both naive endpoints fail

In dimension one, the source's Riesz–Hilbert transform is the classical Hilbert transform below, times a fixed Clifford generator. That generator does not affect integrability or boundedness.

Theorem 3 (Exact endpoint obstructions)**.** *There is a scalar bandlimited $f \in L^1(\mathbb{R})$ such that $Hf \notin L^1(\mathbb{R})$, and a scalar bandlimited $g \in L^\infty(\mathbb{R})$ such that $Hg \notin L^\infty(\mathbb{R})$. Consequently the source decomposition $f_\pm = (f \pm Hf)/2$ cannot define the proposed endpoint theory while keeping both boundary pieces in the same endpoint space.*

Proof. Let

$$f(x) = \left(\frac{\sin x}{x}\right)^2, \qquad \widehat{f}(\xi) = \pi\left(1 - \frac{|\xi|}{2}\right)_+.$$

Thus $f \in L^1$ and is supported in frequency on $[-2,2]$. Multiplication by the Hilbert symbol $-i \operatorname{sgn}\xi$ and direct inversion give

$$Hf(x) = \int_0^2 \left(1 - \frac{\xi}{2}\right) \sin(x\xi)\, d\xi = \frac{1}{x} - \frac{\sin(2x)}{2x^2}. \tag{11}$$

Hence $|Hf(x)| \sim 1/|x|$, so $Hf \notin L^1$.

For the other endpoint, put

$$g(x) = \frac{2}{\pi} \int_0^x \frac{\sin s}{s}\, ds.$$

This function is bounded, and $\widehat{g'} = 2\mathbf{1}_{[-1,1]}$; therefore the distributional Fourier support of g is contained in $[-1, 1]$ (a possible residual term is supported at zero). The Hilbert transform commutes with differentiation, and direct inversion of $-2i\,\mathrm{sgn}(\xi)\mathbf{1}_{[-1,1]}$ yields

$$(Hg)'(x) = \frac{2}{\pi} \frac{1 - \cos x}{x}.$$

Thus, up to an additive constant,

$$Hg(x) = \frac{2}{\pi} \int_0^x \frac{1 - \cos s}{s}\, ds = \frac{2}{\pi} \log|x| + O(1), \tag{12}$$

which is unbounded. □

Proposition 4 (Bandpass endpoint repair)**.** *Fix $0 < \rho < R$. On data with* $\operatorname{supp} \widehat{f} \subseteq \{\rho \le |\xi| \le R\}$*, the Riesz–Hilbert transform is bounded on every L^p, $1 \le p \le \infty$.*

Proof. Choose $\eta \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$ equal to one on the stated annulus. Then

$$Hf = K * f, \qquad K = \mathcal{F}^{-1}\left[\frac{-i\xi}{|\xi|}\eta(\xi)\right] \in \mathcal{S}(\mathbb{R}^n).$$

Young's inequality gives $\|Hf\|_p \le \|K\|_1 \|f\|_p$ at both endpoints and all intermediate exponents. Hence the source Hardy splitting is well-defined on this bandpass subspace. □

5 Verification, limitations, and novelty

The verifier checks the exact Parseval constant on $f = (\sin x/x)^2$, tensor sinc convergence, propagation through a fractional Hardy multiplier, the uniform tail inequality, and the logarithmic slopes in (11) and (12). All tests pass. These computations audit constants; the proofs are analytic.

Limitations. The cube formula oversamples a ball but is exact. We do not prove a general L^p sampling theorem, nor do we replace L^1 by H^1 or L^∞ by BMO. Proposition 4 is a clean endpoint subspace, not a full endpoint Paley–Wiener theorem.

Bounded novelty audit. Cheap run-index searches and focused primary-source searches through 13 August 2026 found no later Riesz–Feller sampling/RKHS follow-up. The cited 2007 paper already samples Clifford-valued ball-bandlimited boundary functions with Bessel expansions [2]. We therefore claim only the explicit operator-valued RKHS, stable boundary-to-spacetime propagation, and the exact endpoint obstruction/repair in the source's new multiplier setting.

Human-review recommendation: send to an expert in sampling theory and Clifford harmonic analysis. Check multiplier handedness, the adjoint in (4), and whether the broad source outlook warrants a "substantial partial" rather than a stronger status.

References

[1] S. Bernstein and N. Faustino, *Paley–Wiener Type Theorems Associated to Dirac Operators of Riesz–Feller Type*, arXiv:2405.04989; J. Fourier Anal. Appl. **31** (2025), article 52.

[2] K. I. Kou, T. Qian, and F. Sommen, *Sampling with Bessel Functions*, Adv. Appl. Clifford Algebras **17** (2007), 519–536.